

HALF/FULL ADDER & HALF/FULL SUBTRACTOR

Aim:

To realize half/full adder and half/full subtractor.

- i. Using X-OR and basic gates.
- ii. Using only NAND gates.

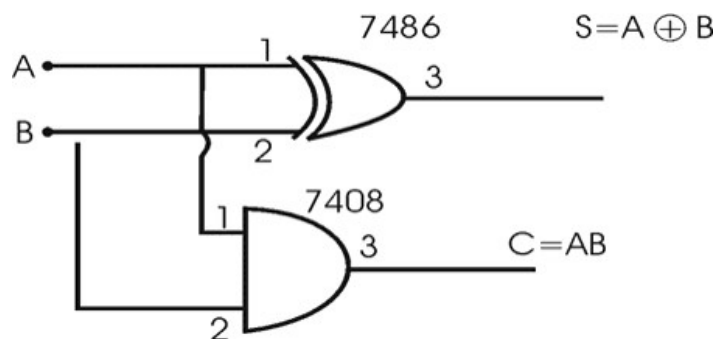
Apparatus Required:

Digital Trainer Kit, IC 7486, IC 7432, IC 7408, IC 7400, and connecting wires

Procedure:

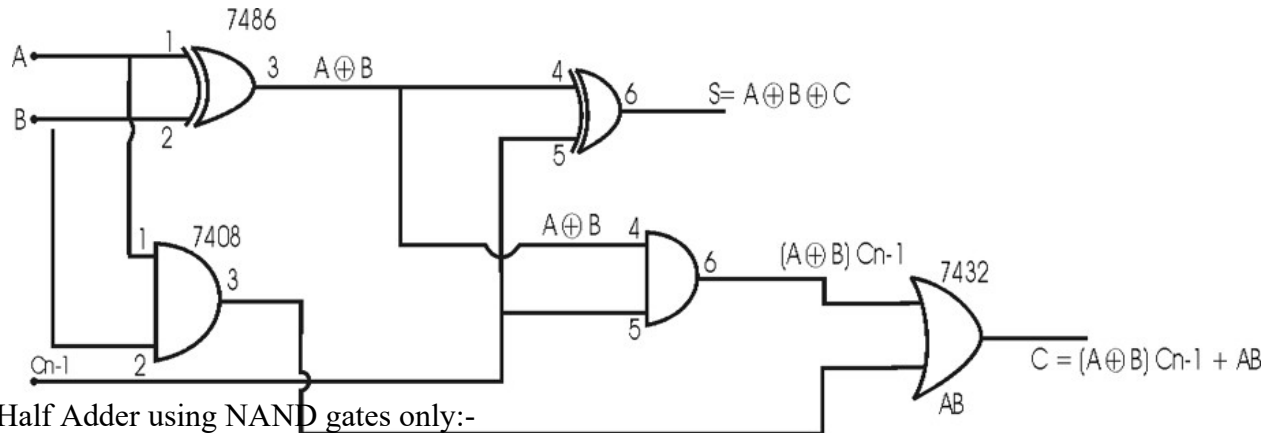
1. Verify the gates.
2. Make the connections as per the circuit diagram.
3. Switch on V_{CC} and apply various combinations of input according to the truth table.
4. Note down the output readings for half/full adder and half/full subtractor sum/difference and the carry/borrow bit for different combinations of inputs.

Half Adder using basic gates:-

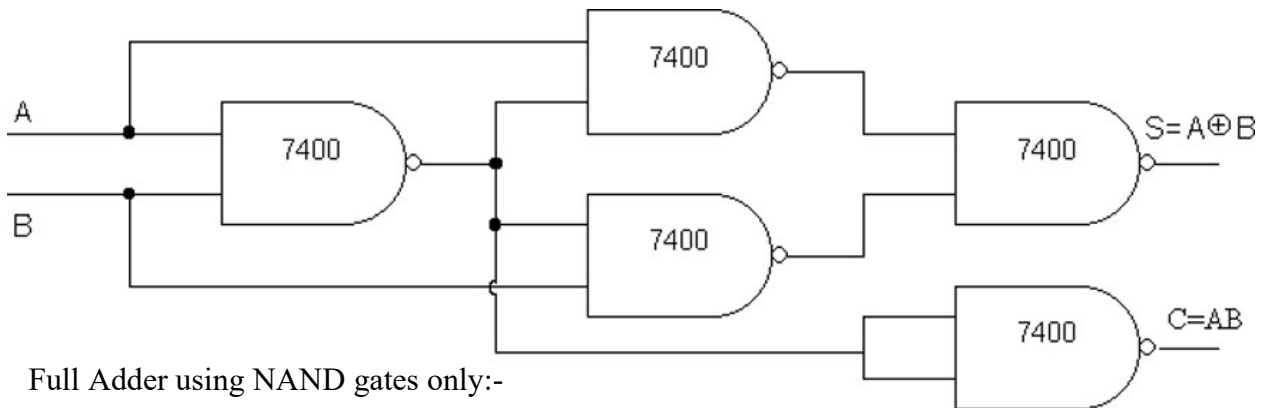


$$S = \bar{A}B + A\bar{B}$$
$$S = A \oplus B$$
$$C = AB$$

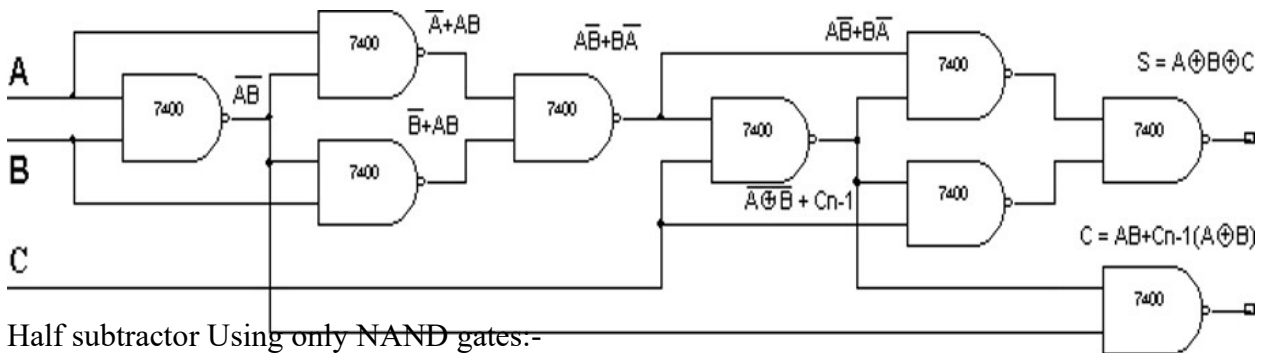
Full Adder using basic gates:-



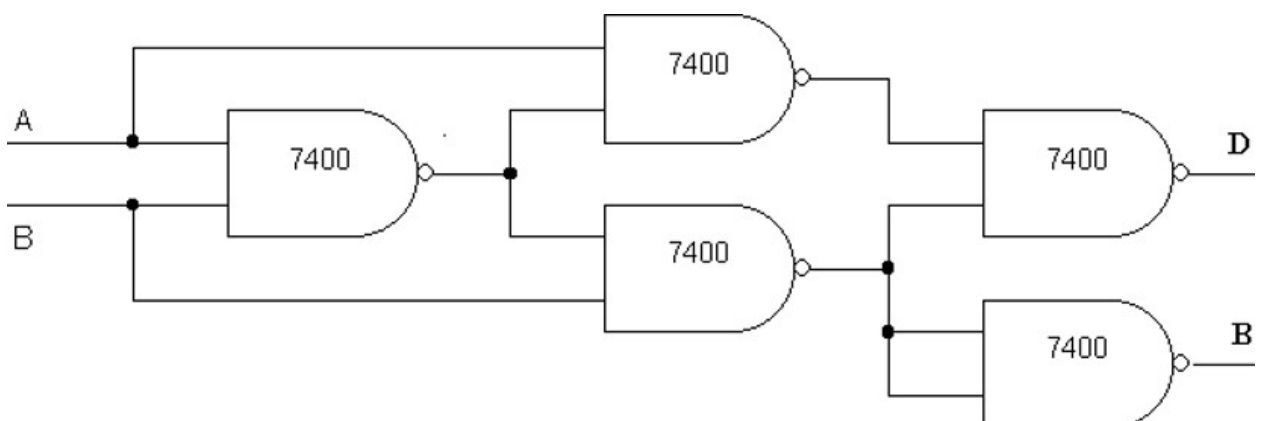
Half Adder using NAND gates only:-



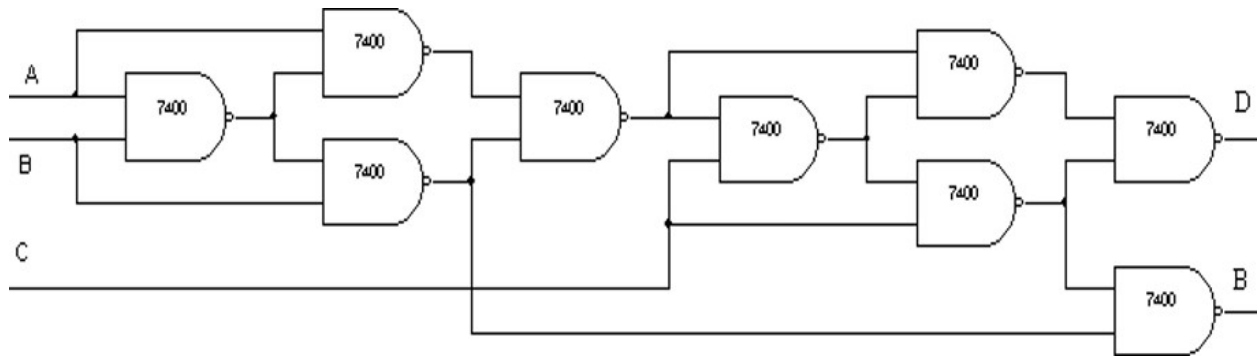
Full Adder using NAND gates only:-



Half subtractor Using only NAND gates:-



Full Subtractor Using only NAND gates:-



Half Adder					
A	B	S	C	S(V)	C(V)
	0	0	0	0	0
0	1	1	0	5	0
1	0	1	0	5	0
1	1	0	1	0	5

Full Adder						
A	B	C _{n-1}	S	C	S(V)	C(V)
0	0	0	0	0	0	0
0	0	1	1	0	5	0
0	1	0	1	0	5	0
0	1	1	0	1	0	5
1	0	0	1	0	5	0
1	0	1	0	1	0	5
1	1	0	0	1	0	5
1	1	1	1	1	5	5

Half Subtractor					
A	B	D	B	D(V)	B(V)
0	0	0	0	0	0
0	1	1	1	5	5
1	0	1	0	5	0
1	1	0	0	0	0

Full Subtractor						
A	B	Cn-1	D	B	D(v)	B(v)
0	0	0	0	0	0	0
0	0	1	1	1	5	5
0	1	0	1	1	5	5
0	1	1	0	1	0	5
1	0	0	1	0	5	0
1	0	1	0	0	0	0
1	1	0	0	0	0	0
1	1	1	1	1	5	5

CONCLUSION:

Thus, the truth tables of Half /Full Adder & Subtractor were verified experimentally and the observed output voltages corresponded to the expected logic levels.